Fiscal Year:	FY 2014	**Task Last Updated:**	FY 04/01/2026
PI Name:	Almeida, Eduardo A C Ph.D.		
Project Title:	The Role of the p53 Pathway in Spaceflight-Induced Tissue Degeneration		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	08/01/2009	**End Date:**	09/30/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Taylor, Elizabeth	**Contact Phone:**	650.604.1783
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Flight Program:	Bion/Shuttle		
Flight Assignment:	STS-135 ; Bion M1 NOTE: Extended to 9/30/2014 per A. Chu/ARC (Ed., 4/3/2014) NOTE: Extended to 7/31/2014 per A. Chu/ARC (Ed., 4/30/2013)		
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Globus, Ruth (NASA Ames Research Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Our working hypothesis is that spaceflight factors (including radiation and microgravity) affect the growth and regenerative health of a broad range of mammalian tissues, including bone, muscle, and immune system, by causing indiscriminate slowing or arrest of normal cell proliferation. Specifically we hypothesize that there is a key regulatory cell-signaling pathway responsible for this space-induced arrest, namely the p53/p21 signaling pathway, and that either radiation and/or microgravity can activate it. We have already tested this hypothesis with spaceflight tissues from STS-131, validating some of its key aspects such as p21activation in osteoblasts, and are now utilizing mouse tissues from the recently completed Bion M1 30-day spaceflight to test the p53 hypothesis with long-term spaceflight tissue samples. Specifically we are working towards determining if the p53/p21 pathway is also altered during 30-day spaceflight in bone tissues by: 1) By measuring gene expression and protein expression and modifications of key signaling molecules that control cell cycle arrest and proliferation such as those in the p53/p21 and PI3Kinase pathways 2) Immunohistochemically identifying specific cell types in bone with active p53/p21 3) Correlating p53/p21 pathway and related cell cycle status to bone loss measurements by histomorphometry, microCT and transmission R-ray microscopy.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The significance of the studies already completed and still underway is several-fold, but first and foremost they are a the test of an original and novel hypothesis that has the potential to be a common element of the already known tissue degenerative conditions observed in space, and an important predictor of yet unknown degenerative conditions that may emerge in long-term spaceflight, specially outside of LEO. There is already incontrovertible ground evidence that the p53 signaling pathway is key in the response to radiation, and there is also compelling evidence that p53 is required for tissue degenerative conditions caused by mechanical unloading. Although spaceflight evidence is limited, skin and muscle in rats flown in space also show elevated p53, strongly compelling us to follow this line of inquiry. The fact that p53 pathway is a powerful tumor suppressor, and inducer of cell cycle arrest (through p21) and programmed cell death, or apoptosis (through p21), should serve as clear indicator if spaceflight factors activate the p53/p21 pathway, this may have systemic effects on all the body processes involving cell proliferation. These processes are most notably the indispensable tissue regeneration that keep every organ from collapsing due to unrepaired cellular and tissue damage. In addition these studies are also highly significant because they attempt to pinpoint the cellular specificity of the p53/p21 pathway activation in bone, and to determine if this is a organ-wide process or limited to certain cell types such as the bone marrow osteoprecursor somatic stem cell lineage. If the p53/p21 pathway is proven to be significantly involved in spaceflight arrest of tissue regeneration and degenerative conditions, it will offer both a new pharmacological target for intervention by drugs such as pifithrin class inhibitors and as a biomarker of tissue damage in space. Finally the Bion M1 studies we are conducting will shed new light on the old question of bone loss in space, possibly extending our understanding of bone dynamics in adult animals, and by being the longest experimental rodent exposure to spaceflight to date. Already our recently published results on bone dynamics during STS-131 show that both osteoclasts, osteoblasts, and osteocytes are involved in bone loss in response to spaceflight, and that both p21and p53 are altered. When the ongoing Bion M1 tissue and cell studies are concluded we expect that we will have gained new factual understanding of bone loss in space under conditions previously not attempted, and will have an important new perspective about the potential for activation of the p53/p21 pathway to be a serious problem for human tissue regeneration during long-term spaceflight, and in human health and disease related to disuse.
Task Progress:	No final report received. Please see the Cumulative Bibliography for published results from this study (Ed.)
Bibliography Type:	Description: (Last Updated: 12/08/2025)
Articles in Peer-reviewed Journals	Blaber EA, Dvorochkin N, Torres ML, Yousuf R, Burns BP, Globus RK, Almeida EA. "Mechanical unloading of bone in microgravity reduces mesenchymal and hematopoietic stem cell-mediated tissue regeneration." Stem Cell Res. 2014 Sep;13(2):181-201. Epub 2014 Jun 9. http://dx.doi.org/10.1016/j.scr.2014.05.005 ; PubMed PMID: 25011075 , Sep-2014